

Block RAM Primitives

The iCE architecture supports dual ported synchronous RAM, with 4096 bits, and a fixed 16 bit data-width. The block is arranged as 256 x 16 bit words. The RAM block may be configured to be used as a RAM with data between 1-16 bits.

iCE40 Block RAM

Each iCE40 device includes multiple high-speed synchronous RAM blocks, each 4Kbit in size. The RAM block has separate write and read ports, each with independent control signals. Each RAM block can be configured into a RAM block of size 256x16, 512x8, 1024x4 or 2048x2. The data contents of the RAM block are optionally pre-loaded during ICE device configuration.

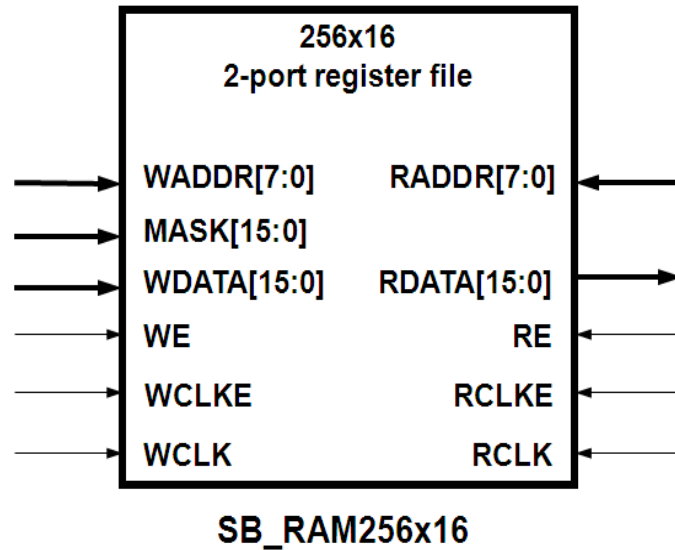
The following table lists the supported dual port synchronous RAM configurations, each of 4Kbits in size. The RAM blocks can be directly instantiated in the top module and taken through iCube2 flow.

Block RAM Configuration	Block RAM Size	WADDR Port Size (Bits)	WDATA Port Size (Bits)	RADDR Port Size (Bits)	RDATA Port Size (Bits)	MASK Port Size (Bits)
SB_RAM256x16 SB_RAM256x16NR SB_RAM256x16NW SB_RAM256x16NRNW	256x16 (4K)	8 [7:0]	16 [15:0]	8 [7:0]	16 [15:0]	16 [15:0]
SB_RAM512x8 SB_RAM512x8NR SB_RAM512x8NW SB_RAM512x8NRNW	512x8 (4K)	9 [8:0]	8 [7:0]	8 [8:0]	8 [7:0]	No Mask Port
SB_RAM1024x4 SB_RAM1024x4NR SB_RAM1024x4NW SB_RAM1024x4NRNW	1024x4 (4K)	10 [9:0]	4 [3:0]	10 [9:0]	4 [3:0]	No Mask Port
SB_RAM2048x2 SB_RAM2048x2NR SB_RAM2048x2NW SB_RAM2048x2NRNW	2048x2 (4K)	11 [10:0]	2 [1:0]	10 [9:0]	2 [1:0]	No Mask Port

The Lattice Technologies convention for the iCE40 RAM primitives with negedge Read or Write clock is that the base primitive name is post fixed with N and R or W according to the clock that is affected, as displayed in the table below for 256x16 RAM block configuration.

RAM Primitive Name	Description
SB_RAM256x16	Posedge Read clock, Posedge Write clock
SB_RAM4256x16NR	Negedge Read clock, Posedge Write clock
SB_RAM256x16NW	Posedge Read clock, Negedge Write clock
SB_RAM256x16NRNW	Negedge Read clock, Negedge Write clock

SB_RAM256x16



The following modules are the complete list of SB_RAM256x16 based primitives

SB_RAM256x16

SB_RAM256x16 //Posedge clock RCLK WCLK
(RDATA, RCLK, RCLKE, RE, RADDR, WCLK, WCLKE, WE, WADDR, MASK, WDATA);

Verilog Instantiation:

```
SB_RAM256x16 ram256x16_inst (  
    .RDATA(RDATA_c[15:0]),  
    .RADDR(RADDR_c[7:0]),  
    .RCLK(RCLK_c),  
    .RCLKE(RCLKE_c),  
    .RE(RE_c),  
    .WADDR(WADDR_c[7:0]),  
    .WCLK(WCLK_c),  
    .WCLKE(WCLKE_c),  
    .WDATA(WDATA_c[15:0]),  
    .WE(WE_c),  
    .MASK(MASK_c[15:0])  
);  
defparam ram256x16_inst.INIT_0 =  
256'h0000000000000000000000000000000000000000000000000000000000000000;  
defparam ram256x16_inst.INIT_1 =  
256'h0000000000000000000000000000000000000000000000000000000000000000;  
defparam ram256x16_inst.INIT_2 =  
256'h0000000000000000000000000000000000000000000000000000000000000000;  
defparam ram256x16_inst.INIT_3 =  
256'h0000000000000000000000000000000000000000000000000000000000000000;  
defparam ram256x16_inst.INIT_4 =  
256'h0000000000000000000000000000000000000000000000000000000000000000;  
defparam ram256x16_inst.INIT_5 =  
256'h0000000000000000000000000000000000000000000000000000000000000000;
```

```

defparam ram256x16_inst.INIT_6 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram256x16_inst.INIT_7 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram256x16_inst.INIT_8 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram256x16_inst.INIT_9 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram256x16_inst.INIT_A =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram256x16_inst.INIT_B =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram256x16_inst.INIT_C =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram256x16_inst.INIT_D =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram256x16_inst.INIT_E =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram256x16_inst.INIT_F =
256'h0000000000000000000000000000000000000000000000000000000000000000;

```

VHDL Instantiation:

```

ram256x16_inst : SB_RAM256x16
generic map (
INIT_0 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_1 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_2 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_3 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_4 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_5 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_6 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_7 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_8 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_9 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_A =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_B =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_C =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_D =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_E =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_F => X"0000000000000000000000000000000000000000000000000000000000000000"
)
port map (
    RDATA => RDATA_C,
    RADDR => RADDR_C,
    RCLK => RCLK_C,
    RCLKE => RCLKE_C,
    RE => RE_C,

```

```

WADDR => WADDR_c,
WCLK=> WCLK_c,
WCLKE => WCLKE_c,
WDATA => WDATA_c,
MASK  => MASK_c,
WE => WE_c
);

```

SB_RAM256x16NR

SB_RAM256x16NR // Negative edged Read Clock – i.e. RCLKN
(RDATA, RCLKN, RCLKE, RE, RADDR, WCLK, WCLKE, WE, WADDR, MASK, WDATA);

Verilog Instantiation:

```

SB_RAM256x16NR ram256x16nr_inst (
    .RDATA(RDATA_c[15:0]),
    .RADDR(RADDR_c[7:0]),
    .RCLKN(RCLKN_c),
    .RCLKE(RCLKE_c),
    .RE(RE_c),
    .WADDR(WADDR_c[7:0]),
    .WCLK(WCLK_c),
    .WCLKE(WCLKE_c),
    .WDATA(WDATA_c[15:0]),
    .WE(WE_c),
    .MASK(MASK_c[15:0])
);
defparam ram256x16nr_inst.INIT_0 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram256x16nr_inst.INIT_1 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram256x16nr_inst.INIT_2 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram256x16nr_inst.INIT_3 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram256x16nr_inst.INIT_4 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram256x16nr_inst.INIT_5 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram256x16nr_inst.INIT_6 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram256x16nr_inst.INIT_7 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram256x16nr_inst.INIT_8 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram256x16nr_inst.INIT_9 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram256x16nr_inst.INIT_A =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram256x16nr_inst.INIT_B =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram256x16nr_inst.INIT_C =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram256x16nr_inst.INIT_D =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram256x16nr_inst.INIT_E =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram256x16nr_inst.INIT_F =
256'h0000000000000000000000000000000000000000000000000000000000000000;

```

VHDL Instantiation:

```
ram256x16nr_inst : SB_RAM256x16NR
generic map (
  INIT_0 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
  INIT_1 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
  INIT_2 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
  INIT_3 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
  INIT_4 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
  INIT_5 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
  INIT_6 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
  INIT_7 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
  INIT_8 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
  INIT_9 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
  INIT_A =>
X"0000000000000000000000000000000000000000000000000000000000000000",
  INIT_B =>
X"0000000000000000000000000000000000000000000000000000000000000000",
  INIT_C =>
X"0000000000000000000000000000000000000000000000000000000000000000",
  INIT_D =>
X"0000000000000000000000000000000000000000000000000000000000000000",
  INIT_E =>
X"0000000000000000000000000000000000000000000000000000000000000000",
  INIT_F => X"0000000000000000000000000000000000000000000000000000000000000000"
)
port map (
  RDATA => RDATA_C,
  RADDR => RADDR_C,
  RCLKN => RCLKN_C,
  RCLKE => RCLKE_C,
  RE => RE_C,
  WADDR => WADDR_C,
  WCLK=> WCLK_C,
  WCLKE => WCLKE_C,
  WDATA => WDATA_C,
  MASK => MASK_C,
  WE => WE_C
);
```

SB_RAM256x16NW

SB_RAM256x16NW **// Negative edged Write Clock – i.e. WCLKN**
(RDATA, RCLK, RCLKE, RE, RADDR, WCLKN, WCLKE, WE, WADDR, MASK, WDATA);

Verilog Instantiation:

```
SB_RAM256x16NW    ram256x16nw_inst (
    .RDATA(RDATA_c[15:0]),
    .RADDR(RADDR_c[7:0]),
    .RCLK(RCLK_c),
    .RCLKE(RCLKE_c),
    .RE(RE_c),
    .WADDR(WADDR_c[7:0]),
    .WCLKN(WCLKN_c),
    .WCLKE(WCLKE_c),
    .WDATA(WDATA_c[15:0]),
    .WE(WE_c),
    .MASK(MASK_c[15:0])
);
defparam ram256x16nw_inst.INIT_0 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram256x16nw_inst.INIT_1 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram256x16nw_inst.INIT_2 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram256x16nw_inst.INIT_3 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram256x16nw_inst.INIT_4 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram256x16nw_inst.INIT_5 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram256x16nw_inst.INIT_6 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram256x16nw_inst.INIT_7 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram256x16nw_inst.INIT_8 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram256x16nw_inst.INIT_9 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram256x16nw_inst.INIT_A =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram256x16nw_inst.INIT_B =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram256x16nw_inst.INIT_C =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram256x16nw_inst.INIT_D =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram256x16nw_inst.INIT_E =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram256x16nw_inst.INIT_F =
256'h0000000000000000000000000000000000000000000000000000000000000000;
```

VHDL Instantiation:

```
ram256x16nw_inst : SB_RAM256x16NW
generic map (
    INIT_0 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
    INIT_1 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
    INIT_2 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
    INIT_3 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
    INIT_4 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
```

```

INIT_5 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_6 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_7 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_8 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_9 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_A =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_B =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_C =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_D =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_E =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_F => X"0000000000000000000000000000000000000000000000000000000000000000"
)
port map (
    RDATA => RDATA_c,
    RADDR => RADDR_c,
    RCLK => RCLK_c,
    RCLKE => RCLKE_c,
    RE => RE_c,
    WADDR => WADDR_c,
    WCLKN=> WCLKN_c,
    WCLKE => WCLKE_c,
    WDATA => WDATA_c,
    MASK => MASK_c,
    WE => WE_c
);

```

SB_RAM256x16NRNW

SB_RAM256x16NRNW // Negative edged Read and Write – i.e. RCLKN WRCKLN
(RDATA, RCLKN, RCLKE, RE, RADDR, WCLKN, WCLKE, WE, WADDR, MASK, WDATA);

Verilog Instantiation:

```

SB_RAM256x16NRNW ram256x16nrnw_inst (
    .RDATA(RDATA_c[15:0]),
    .RADDR(RADDR_c[7:0]),
    .RCLKN(RCLKN_c),
    .RCLKE(RCLKE_c),
    .RE(RE_c),
    .WADDR(WADDR_c[7:0]),
    .WCLKN(WCLKN_c),
    .WCLKE(WCLKE_c),
    .WDATA(WDATA_c[15:0]),
    .WE(WE_c),
    .MASK(MASK_c[15:0])
);
defparam ram256x16nrnw_inst.INIT_0 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram256x16nrnw_inst.INIT_1 =
256'h0000000000000000000000000000000000000000000000000000000000000000;

```

```

defparam ram256x16nrnw_inst.INIT_2 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram256x16nrnw_inst.INIT_3 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram256x16nrnw_inst.INIT_4 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram256x16nrnw_inst.INIT_5 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram256x16nrnw_inst.INIT_6 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram256x16nrnw_inst.INIT_7 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram256x16nrnw_inst.INIT_8 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram256x16nrnw_inst.INIT_9 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram256x16nrnw_inst.INIT_A =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram256x16nrnw_inst.INIT_B =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram256x16nrnw_inst.INIT_C =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram256x16nrnw_inst.INIT_D =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram256x16nrnw_inst.INIT_E =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram256x16nrnw_inst.INIT_F =
256'h0000000000000000000000000000000000000000000000000000000000000000;

```

VHDL Instantiation:

```

ram256x16nrnw_inst : SB_RAM256x16NRNW
generic map (
INIT_0 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_1 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_2 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_3 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_4 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_5 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_6 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_7 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_8 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_9 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_A =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_B =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_C =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_D =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_E =>
X"0000000000000000000000000000000000000000000000000000000000000000",

```

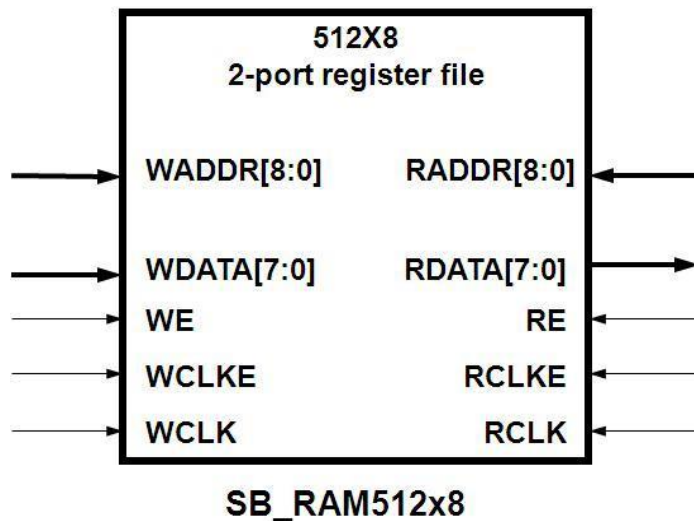


```

INIT_F => X"0000000000000000000000000000000000000000000000000000000000000000"
)
port map (
    RDATA => RDATA_C,
    RADDR => RADDR_C,
    RCLKN => RCLKN_C,
    RCLKE => RCLKE_C,
    RE => RE_C,
    WADDR => WADDR_C,
    WCLKN=> WCLKN_C,
    WCLKE => WCLKE_C,
    WDATA => WDATA_C,
    MASK  => MASK_C,
    WE => WE_C
);

```

SB_RAM512x8



The following modules are the complete list of SB_RAM512x8 based primitives

SB_RAM512x8

SB_RAM512x8 //Posedge clock RCLK WCLK
(RDATA, RCLK, RCLKE, RE, RADDR, WCLK, WCLKE, WE, WADDR, MASK, WDATA);

Verilog Instantiation:

```
SB_RAM512x8 ram512x8_inst (
    .RDATA(RDATA_c[7:0]),
    .RADDR(RADDR_c[8:0]),
    .RCLK(RCLK_c),
    .RCLKE(RCLKE_c),
    .RE(RE_c),
    .WADDR(WADDR_c[8:0]),
    .WCLK(WCLK_c),
    .WCLKE(WCLKE_c),
    .WDATA(WDATA_c[7:0]),
    .WE(WE_c)
);

defparam ram512x8_inst.INIT_0 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram512x8_inst.INIT_1 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram512x8_inst.INIT_2 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram512x8_inst.INIT_3 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram512x8_inst.INIT_4 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram512x8_inst.INIT_5 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
```